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MINI PROJECT: DEVELOPMENT PROGRESS BITM 2113 (WEB APPLICATION DEVELOPMENT)		
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Project Title	FOOD DELIVERY MANAGEMENT FOR GRAB/FOODPANDA	
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Due date: 26 JANUARY 2026		Course : 3 BITM

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1.0 INTRODUCTION

1.1 PROJECT OVERVIEW

The Order Management System for GRAB / Foodpanda Runner Orders is a web-based delivery management system designed to assist restaurants in managing food delivery orders efficiently. The system focuses on coordinating delivery tasks between restaurant staff, delivery runners, and customers by providing a structured workflow from order creation until successful delivery.

This system addresses common problems faced by restaurants such as delayed runner assignment, lack of real-time delivery tracking, and poor communication between staff and runners. By digitizing the order assignment and delivery status process, the system improves operational efficiency and service transparency.

1.2 SYSTEM FEATURES

The system includes the following core features:

- User authentication for Customer, Staff, and Runner roles
- Online and walk-in order creation
- Runner assignment interface for staff
- Real-time delivery status updates by runners
- Notification system for staff
- Order tracking for customers
- Order reporting and analytics
- Member management (Add, Update, Delete)

1.3 TECHNOLOGIES USED

Client-Side Technologies

- HTML – Page structure
- CSS – Layout and interface styling

- JavaScript – Form validation and interaction

Server-Side Technologies

- PHP – Backend logic and authentication
- MySQL – Database management

Development Environment

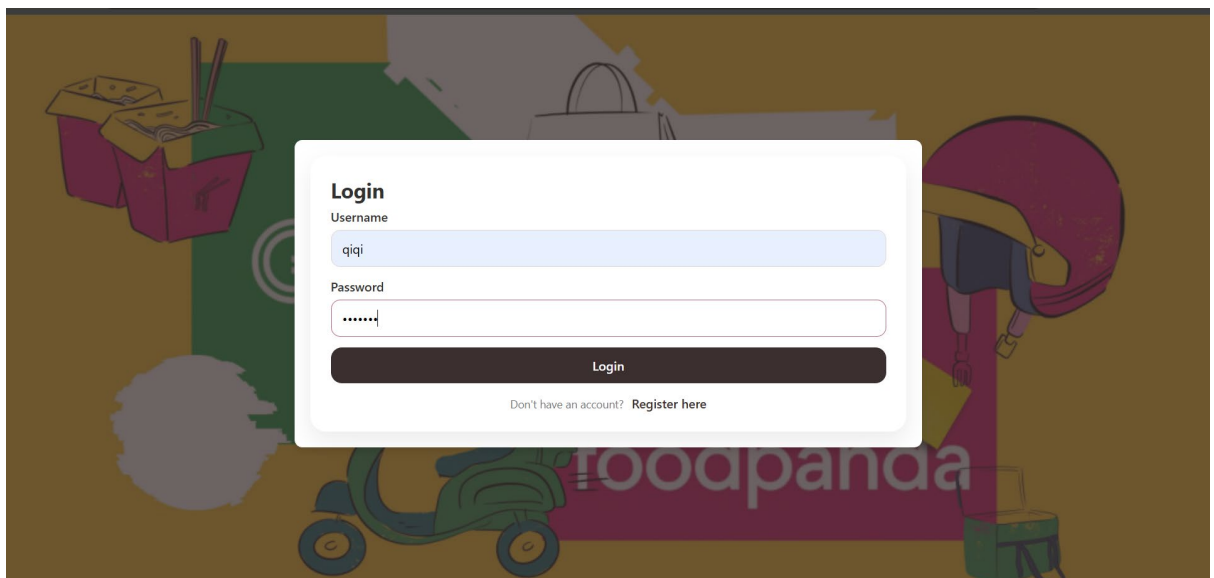
- XAMPP (Apache & MySQL)
- phpMyAdmin
- Web browser for testing

2.0 INTERFACES

2.1 LOG IN PAGE

Purpose:

The Log In Page is used to authenticate users based on their respective roles, which include Customer, Staff, and Runner. The system verifies the username and password entered and manages user access using PHP session handling to ensure secure login.



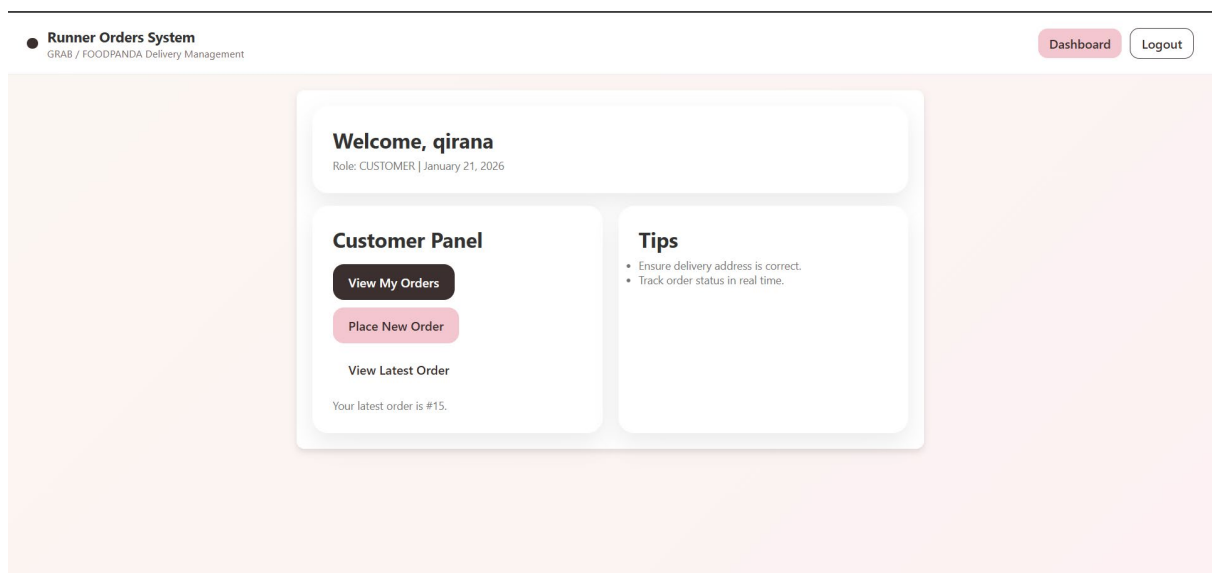
2.2 CUSTOMER PANEL

Purpose:

The Customer Panel allows customers to perform the following actions:

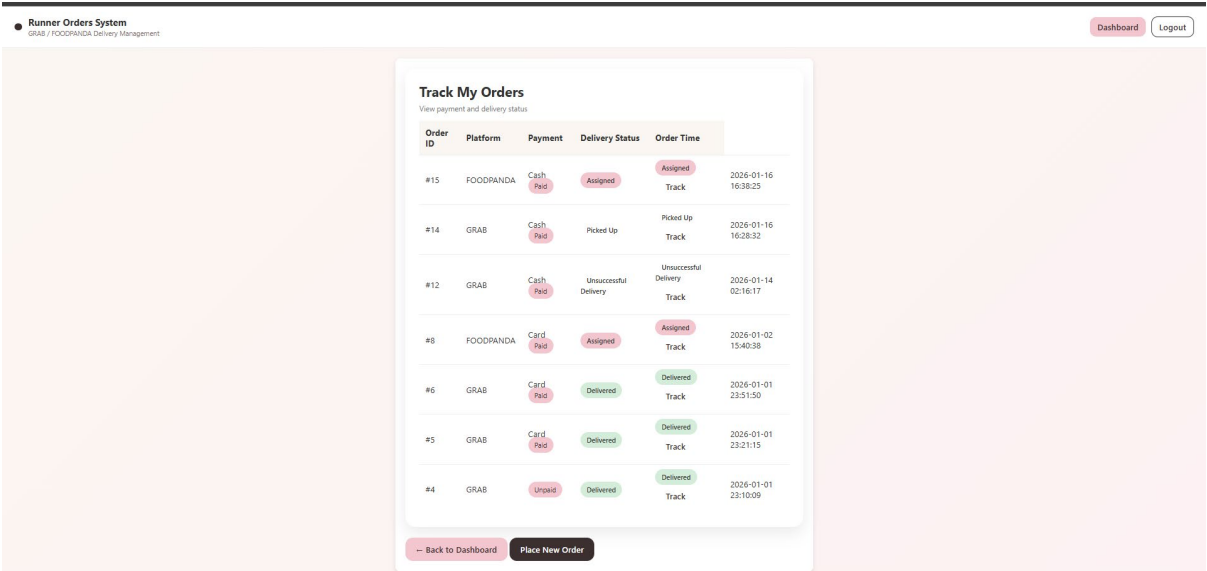
- View dashboard overview
- Place online delivery orders
- View order history
- Track delivery status

Customers are able to monitor their order progress from **Pending** until **Delivered** or **Unsuccessful Delivery**.



2.3 CUSTOMER RUNNER ORDER

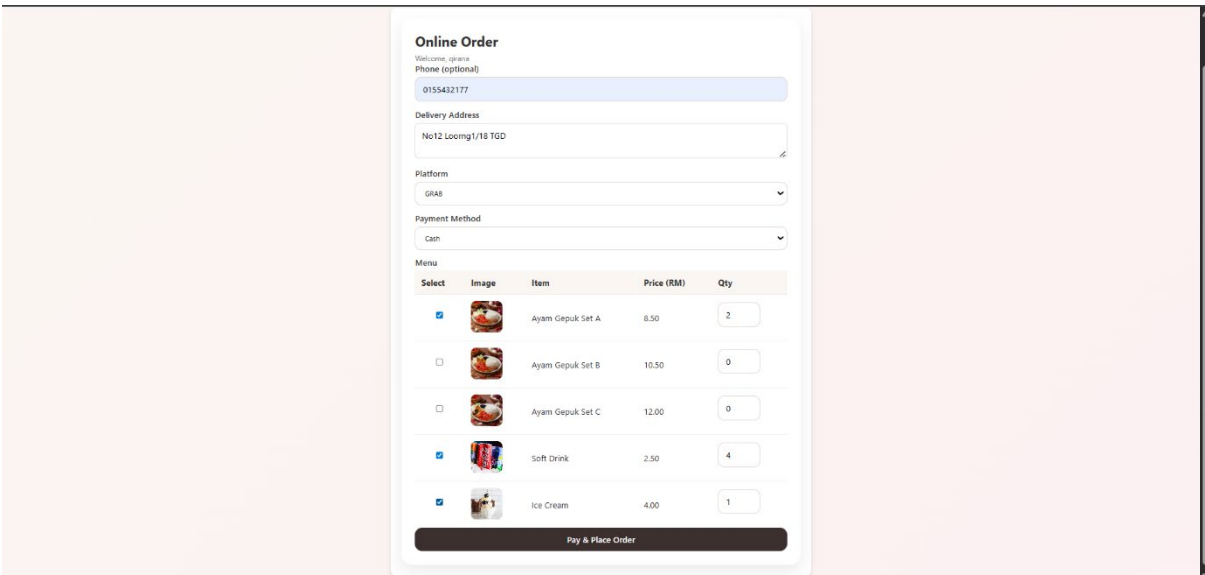
This interface allows customers to view all delivery orders placed using GRAB or Foodpanda. Information such as delivery status and assigned runner details are displayed for tracking purposes.



2.4 CUSTOMER ONLINE ORDER

Customers can place delivery orders by filling in order details including delivery address, selected delivery platform (GRAB or Foodpanda), and selected food items. This interface simplifies the order submission process.

2.5 CUSTOMER VIEW LATEST ORDER



Purpose:

This interface allows customers to:

- View complete order details
- Modify order quantity before confirmation
- Cancel the order if the order status is still pending

View Order #16

Status: Pending

You can modify this order

Customer Name

qirana

Phone Number

0155432177

Delivery Address

No12 Loong1/18 TGD

Platform

GRAB

Order Items

Item	Price (RM)	Quantity	Total (RM)
Ayam Gepuk Set A	8.50	2	17.00
Soft Drink	2.50	4	10.00
Ice Cream	4.00	1	4.00
Grand Total:			RM 31.00

Order Time

2026-01-22 01:28:57

← Back to Dashboard

← View All Orders

Save Changes

Cancel Order

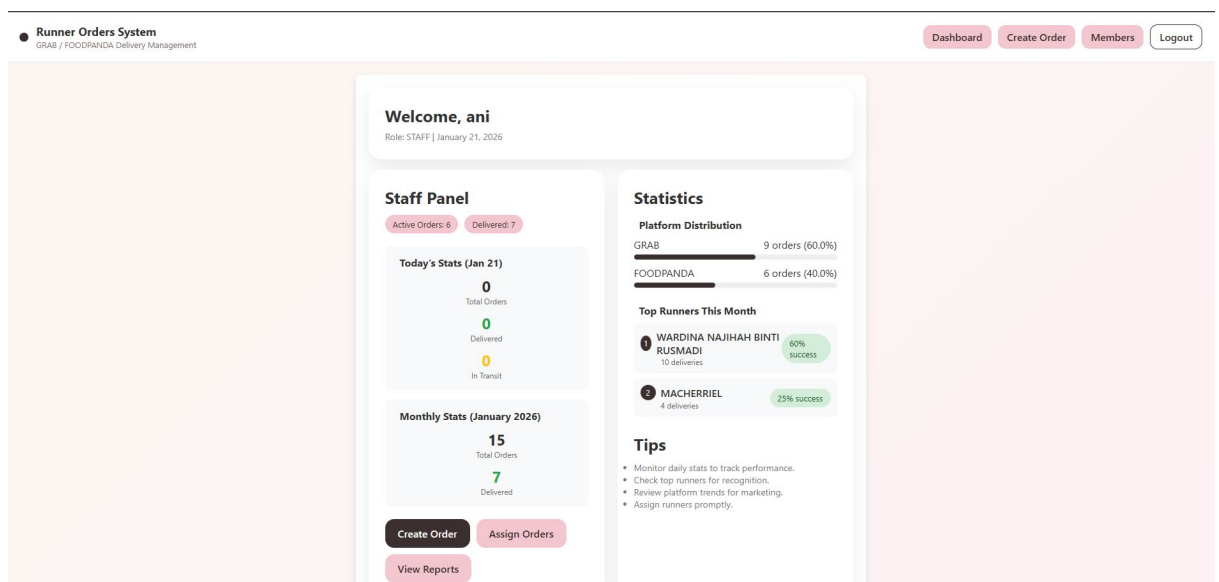
2.6 STAFF PANEL

Purpose:

The Staff Panel functions as the main control centre for restaurant staff. It displays:

- Order statistics
- Platform distribution between GRAB and Foodpanda
- Daily and monthly order summaries

This panel helps staff monitor overall delivery performance.



2.7 STAFF VIEW AND ASSIGN ORDER

Purpose:

This interface allows staff to view detailed order information and assign orders to available delivery runners. It ensures that each delivery order is assigned systematically and efficiently.

Order #16

Platform: GRAB

Created by: Default Staff

Runner: Not Assigned

Pending

Customer Name
qirana

Phone
0155432177

Delivery Address
No12 Loong1/18 TGD

Order Items

Item	Qty	Price (RM)	Total (RM)
Ayam Gepuk Set A	2	8.50	17.00
Soft Drink	4	2.50	10.00
Ice Cream	1	4.00	4.00

Grand Total: RM 31.00

Assign Runner

Click and assign this order to a runner

Select Runner

WARDINA NAJIHAH BINTI RUSMADI

Assign Runner

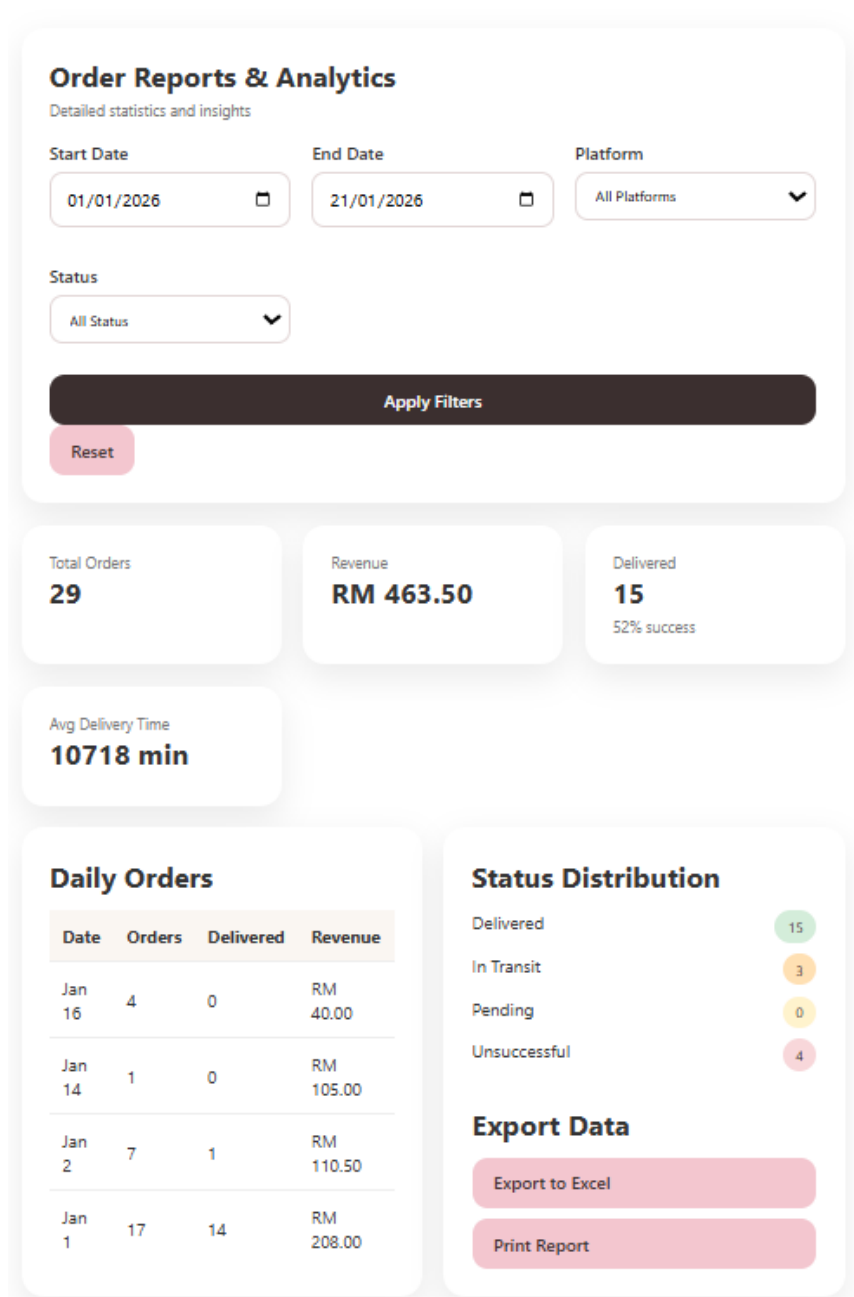
[← Back to Dashboard](#)

2.8 STAFF ADD, UPDATE AND DELETE MEMBER

Staff are able to:

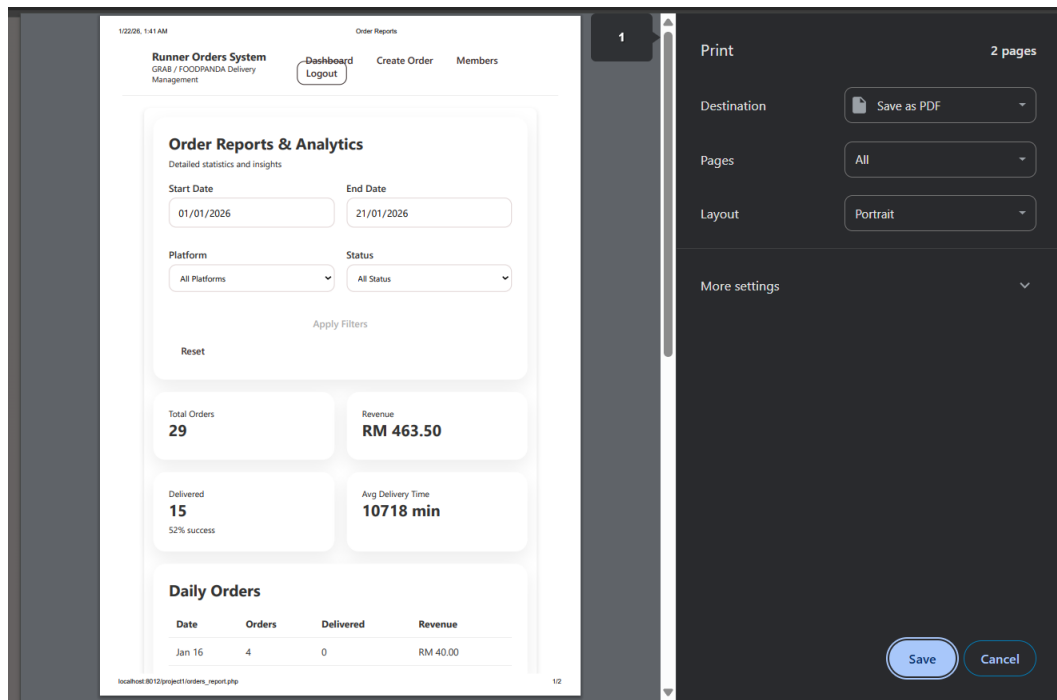
- Add new users
- Update existing member details
- Delete inactive or unnecessary members

This module supports full **CRUD (Create, Read, Update, Delete)** functionality.



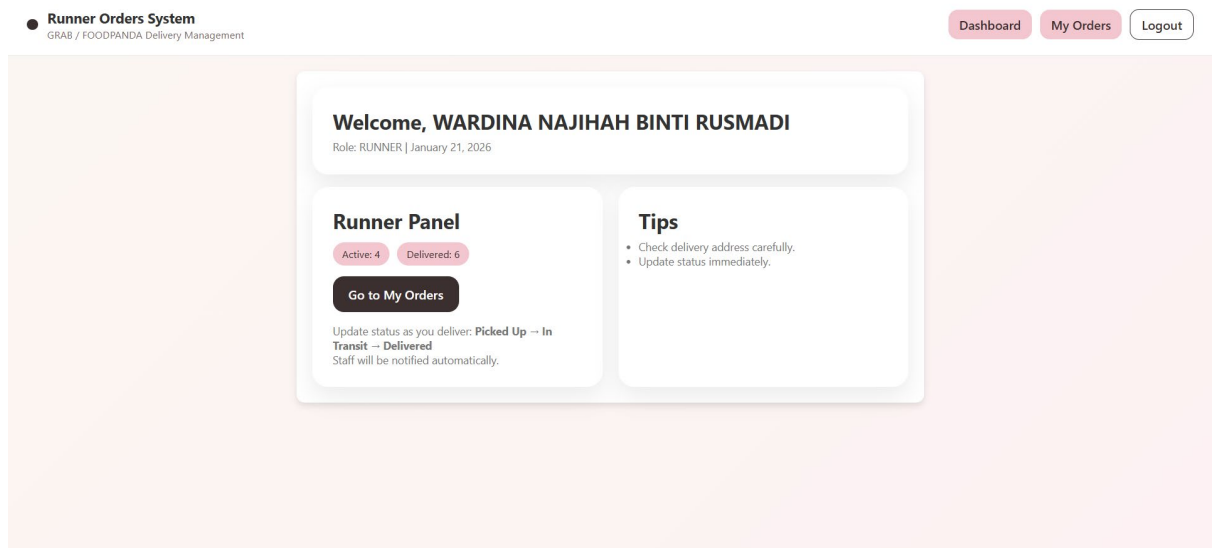
2.9 STAFF PRINT REPORT

This interface allows staff to generate and print delivery reports based on selected criteria such as date range, platform, and order status. The printed reports can be used for record-keeping and performance evaluation.



2.10 RUNNER PANEL

The Runner Panel provides delivery runners with access to their assigned orders. Runners can view customer details, delivery addresses, and current order status.



2.11 RUNNER ASSIGN ORDER PAGE

This page allows runners to update delivery status in real time, such as:

- Assigned
- Picked Up
- In Transit
- Delivered
- Unsuccessful Delivery

Each update is automatically recorded in the system and reflected in the staff panel.

My Assigned Orders

Update status and staff will be notified.

#	Customer	Address	Platform	Status	Update
#15	qirana	no34 lrg gd 2	FOODPANDA	Assigned	Assigned Status updates are saved instantly.
#14	qirana	no35 lorong gd7	GRAB	Delivered	Delivered Status updates are saved instantly.
#12	qirana	16 Jalan Sejahtera	GRAB	In Transit	In Transit Status updates are saved instantly.
#10	Nilam Fazliana	No12 Jalan 3 melaka	FOODPANDA	Unsuccessful Delivery	Unsuccessful Delivery Status updates are saved instantly.

3.0 STYLING USING CSS

3.1 CSS ARCHITECTURE AND MODULAR DESIGN

The system applies a consistent CSS structure to ensure uniform design across all pages.

Modular styling is used to improve maintainability and readability of the CSS code.

3.2 DESIGN PATTERNS

3.2.1 CARD BASED LAYOUT

A card-based layout is implemented to present content in an organised and visually clear manner. This improves user experience and readability.

3.3 RESPONSIVE DESIGN

The system layout is designed to be responsive, ensuring compatibility across different screen sizes such as desktop and mobile devices.

4.0 VALIDATION USING JAVASCRIPT

4.1 CLIENT-SIDE FORM VALIDATION

JavaScript is used to validate user inputs before data is submitted to the server, reducing unnecessary server requests and improving user experience.

4.1.1 LOGIN FORM VALIDATION

The system implements comprehensive login form validation to ensure data integrity:

```
function validateLogin(form) {  
  
    const username = form.username.value.trim();  
  
    const pass = form.password.value;  
  
    let errors = [];
```

```

// Required field validation

if (username.length < 1) errors.push("Enter username.");

if (pass.length < 1) errors.push("Enter password.");


// Empty input prevention

if (username === "" || username === null) errors.push("Username cannot be
empty.");

if (pass === "" || pass === null) errors.push("Password cannot be empty.");


// Display errors or allow submission

showErrors(form, errors);

return errors.length === 0;

}

```

4.1.2 SIGNUP FORM VALIDATION

The registration form includes comprehensive validation for data quality:

```

function validateRegister(form) {

  const role = form.role.value.trim();

  const name = form.name.value.trim();

  const username = form.username.value.trim();

  const email = form.email.value.trim();

  const phone = form.phone.value.trim();

  const pass = form.password.value;

  const pass2 = form.confirm_password.value;

```

```

let errors = [];

// Required field checking

if (!["staff","runner","customer"].includes(role)) errors.push("Role is
invalid.");

if (name.length < 3) errors.push("Name must be at least 3 characters.");

if (username.length < 3) errors.push("Username must be at least 3
characters.");


// Input format validation

if (!isEmail(email)) errors.push("Email format is invalid.");

if (phone && phone.length < 8) errors.push("Phone looks too short.");

if (pass.length < 6) errors.push("Password must be at least 6 characters.");


// Password matching validation

if (pass !== pass2) errors.push("Confirm password does not match.");


showErrors(form, errors);

return errors.length === 0;

}

Email Validation Form:

function isEmail(val) {

    return /^[^s@]+@[^s@]+\.[^s@]+$/.test(val);

}

```

4.2 CHECKOUT FORM VALIDATION

Validation ensures that required information such as delivery address and order details are completed before order submission.

```
// Enhanced form validation for checkout
```

```
function validateCheckoutForm(form) {
```

```
    const errors = [];
```

```
    // Required fields validation
```

```
    const deliveryAddress = form.delivery_address.value.trim();
```

```
    const paymentMethod = form.payment_method.value;
```

```
    if (!deliveryAddress) {
```

```
        errors.push("Delivery address is required.");
```

```
    }
```

```
    if (!paymentMethod) {
```

```
        errors.push("Payment method must be selected.");
```

```
    }
```

```
    // Order items validation
```

```
    let hasItems = false;
```

```
    let totalItems = 0;
```

```
    form.querySelectorAll('input[name^="menu"]').forEach((checkbox, index) => {
```



```
if (checkbox.checked) {  
  
    const qtyInput = form.querySelector(`input[name="qty[${index}]"]`);  
  
    const quantity = parseInt(qtyInput.value) || 0;  
  
    if (quantity > 0) {  
  
        hasItems = true;  
  
        totalItems += quantity;  
  
    }  
  
}  
  
});
```

```
if (!hasItems) {  
  
    errors.push("Please select at least one menu item.");  
  
}
```

```
if (totalItems === 0) {  
  
    errors.push("Please specify quantities for selected items.");  
  
}
```

```
// Phone validation (optional but if provided, must be valid)
```

```
const phone = form.customer_phone.value.trim();  
  
if (phone && !isValidPhone(phone)) {  
  
    errors.push("Phone number format is invalid.");  
  
}
```

```

// Display errors

if (errors.length > 0) {

    showFormErrors(form, errors);

    return false;

}

return true;

}

function isValidPhone(phone) {

    // Basic phone validation for Malaysia

    const phoneRegex = /^(\\+?6?01)[0-46-9]-*[0-9]{7,8}$/;

    return phoneRegex.test(phone.replace(/s+/g, ""));

}

function showFormErrors(form, errors) {

    const errorContainer = form.querySelector('.js-error') || createErrorContainer(form);

    errorContainer.innerHTML = errors.map(error =>

        `<div    class="error-message"><i    class="fas    fa-exclamation-circle"></i>

    ${error}</div>`

    ).join("");

    errorContainer.style.display = 'block';

}

```

4.3 REAL-TIME VALIDATION

4.3.1 EMAIL AVAILABILITY CHECK

Checks whether the email already exists in the database.

// Email availability check with debouncing

```
function setupEmailValidation() {  
  
    const emailInput =  
document.querySelector('input[type="email"][name="email"]');  
  
    if (emailInput) {  
  
        let timeout = null;  
  
        emailInput.addEventListener('input', function() {  
  
            clearTimeout(timeout);  
  
            const email = this.value.trim();  
  
            if (!email || !isEmail(email)) {  
  
                return; // Don't check invalid emails  
  
            }  
  
            timeout = setTimeout(() => {  
  
                checkEmailAvailability(email);  
  
            }, 500); // 500ms debounce  
  
        });  
  
    }  
  
}
```

```
async function checkEmailAvailability(email) {

    const    errorElement    =    document.getElementById('email-error')    ||
createEmailErrorElement();

    try {

        const response = await fetch('api_check_email.php', {

            method: 'POST',

            headers: {

                'Content-Type': 'application/x-www-form-urlencoded',

            },

            body: `email=${encodeURIComponent(email)}`

        });

        const data = await response.json();

        if (data.available === false) {

            errorElement.textContent = 'This email is already registered!';

            errorElement.style.color = '#dc3545';

            errorElement.style.display = 'block';

        } else if (data.available === true) {

            errorElement.textContent = 'Email is available!';

            errorElement.style.color = '#28a745';

            errorElement.style.display = 'block';

        }

    }
```

```
    } catch (error) {  
  
        console.error('Email check failed:', error);  
  
    }  
  
}
```

5.0 WEB SERVER CONFIGURATION

5.1 SERVER INSTALLATION

Apache Web Server is installed using XAMPP.

5.2 APACHE CONFIGURATION

Document Root Configuration

DocumentRoot "C:/xampp/htdocs"

<Directory "C:/xampp/htdocs">

Options Indexes FollowSymLinks Includes ExecCGI

AllowOverride All

Require all granted

</Directory>.

5.3 PHP CONFIGURATION

PHP settings are configured to support session handling and error reporting.

5.4 URL ACCESS

Login Page: <http://localhost:8012/login.php>

6.0 DATABASE DESIGN

6.1 DATABASE OVERVIEW

The database is designed to support user management, order processing, delivery tracking, and notification management

6.2 DATABASE CREATION

A MySQL database is created to store system data such as users, orders, order items, and delivery status records.

CREATE DATABASE runner_orders;

Table	Action	Rows	Type	Collation	Size	Overhead
☐ notifications		25	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ orders		15	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ order_history		0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ order_items		31	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ users		7	InnoDB	utf8mb4_general_ci	48.0 KiB	-
5 tables	Sum	78	InnoDB	utf8mb4_general_ci	224.0 KiB	0 B

Showing rows 0 - 24 (25 total, Query took 0.0003 seconds.)

SELECT * FROM `notifications`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

			id	staff_id	order_id	message	is_read	created_at	
☐				1	1	3	Runner updated status: Assigned → Picked Up	0	2026-01-01 22:41:53
☐				2	3	2	Runner updated status: Assigned → Delivered	1	2026-01-01 22:41:57
☐				3	1	1	Runner updated status: Assigned → Delivered	0	2026-01-01 22:42:15
☐				4	3	7	Runner updated status: Assigned → In Transit	0	2026-01-01 23:55:41
☐				5	1	5	Runner updated status: Assigned → In Transit	0	2026-01-01 23:55:44
☐				6	3	7	Runner updated status: In Transit → Delivered	0	2026-01-01 23:55:50
☐				7	3	7	Runner updated status: Delivered → Picked Up	0	2026-01-02 15:44:51
☐				8	3	7	Runner updated status: Picked Up → In Transit	0	2026-01-02 16:57:38
☐				9	1	5	Runner updated status: In Transit → Delivered	0	2026-01-02 16:57:43
☐				10	3	11	Runner updated status: Assigned → Picked Up	0	2026-01-02 16:58:15
☐				11	1	3	Runner updated status: Picked Up → Delivered	0	2026-01-14 03:48:54
☐				12	1	4	Runner updated status: Assigned → Delivered	0	2026-01-14 03:48:58
☐				13	1	6	Runner updated status: Assigned → Delivered	0	2026-01-14 03:49:00
☐				14	3	9	Runner updated status: Assigned → Delivered	0	2026-01-14 03:49:03

Showing rows 0 - 14 (15 total, Query took 0.0003 seconds.)

SELECT * FROM `orders`

Profiling

Edit inline

Edit

Explain SQL

Create PHP code

Refresh

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

		id	customer_name	customer_phone	delivery_address	platform	status	assigned_runner_id	created_by_staff_id	created_at	updated_at	pa		
☐				1	faz	0155432177	Dewan Seri Makmur Sultan Haji Ahmad Shah Semambu	FOODPANDA	Delivered	2	1	2026-01-01 22:21:03	2026-01-01 22:42:15	NU
☐				2	faz	0143543212	30, Laluan Sakura26, Perak	GRAB	Delivered	2	3	2026-01-01 22:34:13	2026-01-01 22:41:57	NU
☐				3	WARDINA NAJIHAH BINTI RUSMADI	01117975925	NO15 LORONG1/19 TAMAN GAMBANG DAMAI	GRAB	Delivered	2	1	2026-01-01 22:40:23	2026-01-14 03:48:54	NU
☐				4	qirana	0155432177	jalan sakuraa 26	GRAB	Delivered	2	1	2026-01-01 23:10:09	2026-01-14 03:48:58	NU
☐				5	qirana	0155432177	taman tambun perdana	GRAB	Delivered	6	1	2026-01-01 23:21:15	2026-01-02 16:57:43	Ca
☐				6	qirana	0199345422	no 35 lorong gd7	GRAB	Delivered	2	1	2026-01-01 23:51:50	2026-01-14 03:49:00	Ca
☐				7	faz	0199876544	Dewan Seri Makmur Sultan Haji Ahmad Shah	FOODPANDA	In Transit	6	3	2026-01-01 23:54:06	2026-01-02 16:57:38	NU

Server: 127.0.0.1 » Database: runner_orders » Table: order_items

Browse

Structure

SQL

Search

Insert

Export

Import

P

Showing rows 0 - 24 (31 total, Query took 0.0005 seconds.)

SELECT * FROM `order\_items`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1

>

>>

☐ Show all

Number of rows: 25

Filter rows: Search this table

Extra options

				id	order_id	item_name	qty	price
☐	Edit	Copy	Delete	1	1	chicken chop	1	0.00
☐	Edit	Copy	Delete	2	1	fanta grape	1	0.00
☐	Edit	Copy	Delete	3	1	es milo	2	0.00
☐	Edit	Copy	Delete	4	2	Ayam Gepuk Set C	1	12.00
☐	Edit	Copy	Delete	5	2	Soft Drink	1	2.50
☐	Edit	Copy	Delete	6	3	Ayam Gepuk Set A	1	8.50
☐	Edit	Copy	Delete	7	3	Ice Cream	1	4.00
☐	Edit	Copy	Delete	8	4	Ayam Gepuk Set A	1	8.50
☐	Edit	Copy	Delete	9	4	Soft Drink	1	2.50
☐	Edit	Copy	Delete	10	4	Ice Cream	1	4.00
☐	Edit	Copy	Delete	11	5	Ayam Gepuk Set A	1	8.50
☐	Edit	Copy	Delete	12	6	Ayam Gepuk Set C	10	12.00
☐	Edit	Copy	Delete	13	6	Soft Drink	1	2.50
☐	Edit	Copy	Delete	14	6	Ice Cream	5	4.00
☐	Edit	Copy	Delete	15	7	Ayam Gepuk Set A	1	8.50

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Triggers

Showing rows 0 - 6 (7 total, Query took 0.0005 seconds.)

SELECT * FROM `users`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

				id	role	name	username	email	phone	password_hash	created_at
☐	Edit	Copy	Delete	1	staff	Default Staff	staff	staff@example.com	0100000000	\$2y\$10\$wN5YqUjms3yXG8hO1Y0wO1GgP0T8mU8J1zPZqk9Fq...	2026-01-01 21:48:39
☐	Edit	Copy	Delete	2	runner	WARDINA NAJIHAH BINTI RUSMADI	wawa	njhrsmadi@gmail.com	01117975925	\$2y\$10\$4YmVhMg0K7.w4n6F7bNfeTIHRIKdXpquSI.sziCPM...	2026-01-01 22:09:41
☐	Edit	Copy	Delete	3	staff	ani	anini	ani@gmail.com	0188765433	\$2y\$10\$kitVnUhORloqVZdtCOYBduZ29d/OaCzApbKtm7pE4fD...	2026-01-01 22:15:13
☐	Edit	Copy	Delete	5	customer	qirana	qiqi	qiqi@gmail.com	NULL	\$2y\$10\$754eUqotBEyulQpKKOZ8m.JMbbGXsdUBiP8LbmQly...	2026-01-01 22:51:57
☐	Edit	Copy	Delete	6	runner	MACHERRIEL	CHERRY	cherry@gmail.com	NULL	\$2y\$10\$V3ApDN38Lvyh0J7Bhl.YqO3N9r2gt1eR/.78/V3PNzW...	2026-01-01 23:49:24
☐	Edit	Copy	Delete	7	staff	hanie	hanie@gmail.com	0188765456	0188765456	\$2y\$10\$R3o3DevlBlmhUgPwObxb8eQM/7Pdd8SuikQ4nEvHy15...	2026-01-01 23:54:52
☐	Edit	Copy	Delete	8	customer	Nilam Fazliana	Fazliana	Faz@gmail.com	NULL	\$2y\$10\$3zRh9lbiK.cU/9JgUJ0.DJ90ueb9lh5DJIXqudGk...	2026-01-02 16:53:24

☐ Check all With selected: Edit Copy Delete Export